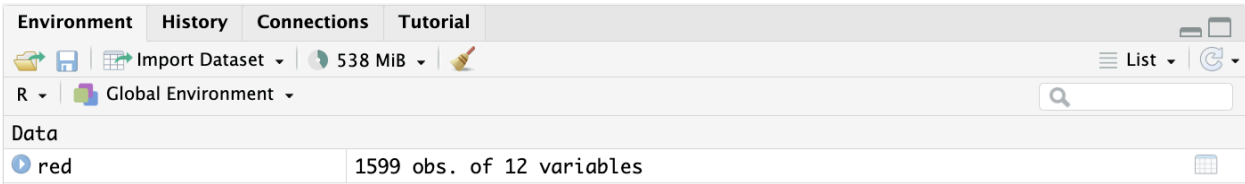
- **What is the sample size?**

The sample size for winequality-red.csv is 1599 wines. We can easily check this by importing our data looking at the environment where our sample size and columns are visible.

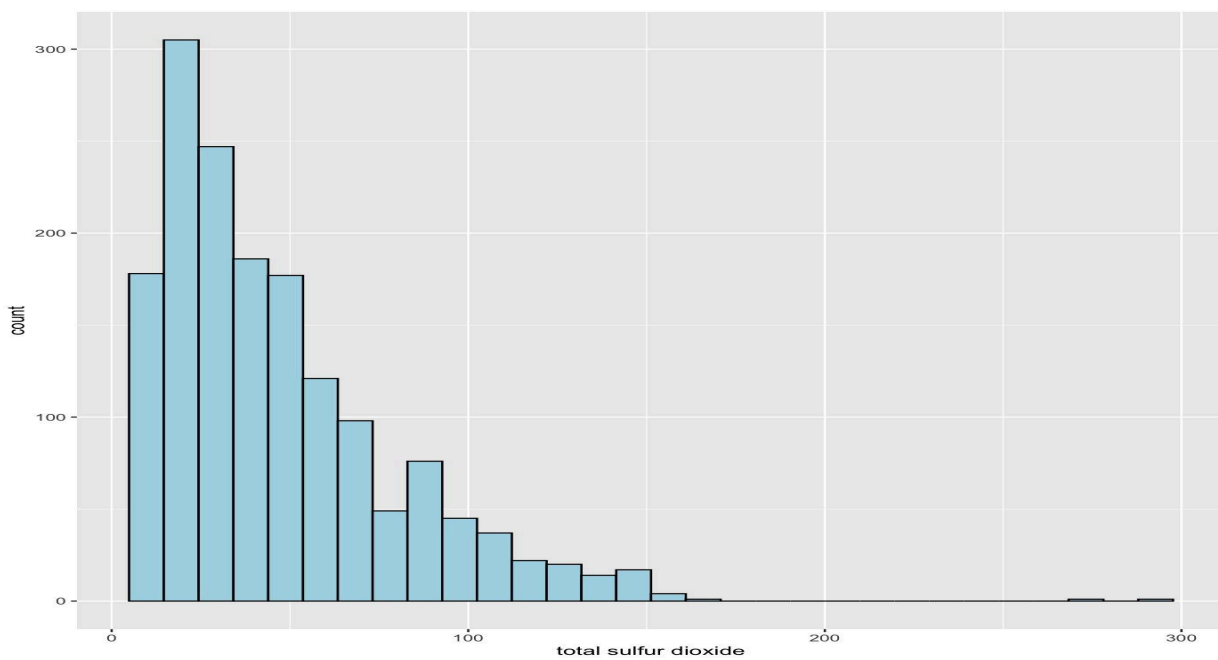
```
red <- read_delim("~/Desktop/Stat Methods/winequality-red.csv", delim = ";")
```



- **Any outliers? Do you have any concerns about the data quality?**

The most noticeable outlier when analyzing the data was in total sulfur dioxide. This was at first concerning because it could mean red wine oxidizes and spoils quickly. After further investigation, it does not affect the mean drastically. When removing the two outliers the mean changed from 46.468 to a 46.171.

```
red %>%
  ggplot() +
  geom_histogram(aes(x = `total sulfur dioxide`),
                 fill = "lightblue",color = "black")
```



- **How can you summarize the data of each variable in a concise way? What statistics are you going to present?**

By using the summary in R Studio, we can easily take simple statistics of each variable in a very concise way. I would start by presenting median and mean; these two statistics, when compared, are a great way

to detect outliers, and understand a balance point of the data. Also, after creating histograms for each variable, showing the audience a visual is easy for reading and understanding the data.

summary(winequality-red.csv)

```
> summary(red)
 fixed acidity   volatile acidity  citric acid    residual sugar     chlorides       free sulfur dioxide total sulfur dioxide    density
 Min.   : 4.60   Min.   :0.1200   Min.   :0.000   Min.   : 0.900   Min.   :0.01200   Min.   : 1.00       Min.   :  6.00       Min.   :0.9901
 1st Qu.: 7.10   1st Qu.:0.3900   1st Qu.:0.090   1st Qu.: 1.900   1st Qu.:0.07000   1st Qu.: 7.00       1st Qu.: 22.00       1st Qu.:0.9956
 Median : 7.90   Median :0.5200   Median :0.260   Median : 2.200   Median :0.07900   Median :14.00       Median : 38.00       Median :0.9968
 Mean   : 8.32   Mean   :0.5278   Mean   :0.271   Mean   : 2.539   Mean   :0.08747   Mean   :15.87       Mean   : 46.47       Mean   :0.9967
 3rd Qu.: 9.20   3rd Qu.:0.6400   3rd Qu.:0.420   3rd Qu.: 2.600   3rd Qu.:0.09000   3rd Qu.:21.00       3rd Qu.: 62.00       3rd Qu.:0.9978
 Max.   :15.90   Max.   :1.5800   Max.   :1.000   Max.   :15.500   Max.   :0.61100   Max.   :72.00       Max.   :289.00       Max.   :1.0037
       pH           sulphates         alcohol         quality
 Min.   :2.740   Min.   :0.3300   Min.   : 8.40   Min.   :3.000
 1st Qu.:3.210   1st Qu.:0.5500   1st Qu.: 9.50   1st Qu.:5.000
 Median :3.310   Median :0.6200   Median :10.20   Median :6.000
 Mean   :3.311   Mean   :0.6581   Mean   :10.42   Mean   :5.636
 3rd Qu.:3.400   3rd Qu.:0.7300   3rd Qu.:11.10   3rd Qu.:6.000
 Max.   :4.010   Max.   :2.0000   Max.   :14.90   Max.   :8.000
```

- **How can you visualize the distribution of each variable?**

Like mentioned above, for showing singular variables, histograms are the best option. Histograms are made for singular and are the most common graph for showing it.

Since a histogram is shown above here is the code and why I chose specific colors.

```
red %>%
  ggplot() +
  geom_histogram(aes(x = `fixed acidity`),
                 fill = "lightblue", color = "black")
```

The geom_histogram function is the basic tidyverse package function that allows for graphs to be created. When creating a histogram it will show dark gray bins (each column); this is fine for a basic graph, but adding color and borders helps with graphical excellence (a standard practice in analyzing). “Fill” is the color of the bins, and “color” is what the borders will be to easily see each bin.

Another way to visualize the variables is combining quality with one to see which correlates the most to a good wine. What I found was alcohol had a positive correlation to quality, whereas volatile acidity related to a negative correlation. A scatter plot is the most common graph for a comparison such as this.

```
cor(red$alcohol, red$quality)
cor(red$`volatile acidity`, red$quality)
```

- **Do you see any skewed distributions?**

Residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, sulphates, alcohol, fixed acidity, and volatile acidity are all variables that I would have skews that are extreme to mild in no particular order. In R studio, if you install the library**(e1071)**, a function called skewness becomes available. This will then calculate a measure of asymmetry. I considered all graphs that were above .5 to count towards my list. Of skewed distributions.

Example: skewness(red$alcohol)